DEVIN PHILLIP DAVIS

Founder, Lead Systems Architect & Quantum Cyberinfrastructure Researcher

Agile Defense Systems LLC · United States

LinkedIn: https://www.linkedin.com/in/ξ-devin-phillip-davis-88379b174 · **Clearance / Status:** CUI / CMMC 2.0 Ready · SAM Active

DAU Acquisition Certifications: CYB 5610, ENG 101/201, WSA 002, ACQ 0150 (MTA), CLC 066 (OTAs), CLM 094 (IMS)

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          Sovereign AI · Low-Level C-SIMD · Quantum Information Science · Cryptographic Governa
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 [ Core Ethos ] Generative models propose actions; mathematical determinism strictly decides exe
 [ Substrates ] x86_64 Linux (WSL2), ARM64 Android 16 (Termux/NDK), Physical IBM Heron r2 QPUs (1
 [ Governance ] SYS-021 Zero Self-Supplied Authorization · FIPS 204 ML-DSA-65 + Ed25519 Hybrid Du
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EXECUTIVE SUMMARY & ARCHITECTURAL ETHOS

Devin Phillip Davis is a Principal Systems Architect, Low-Level Systems Engineer, and Quantum Information Science Researcher with over 8 years of cross-disciplinary experience engineering mathematically verified, fail-closed computing substrates. Davis is the creator and architect of **osiris**, a sovereign computing environment and non-causal language model (NCLM) ecosystem engineered to eliminate cloud reliance, memory stride corruption, and probabilistic model hallucination in mission-critical environments.

His core architectural ethos asserts that **generative models propose actions, while deterministic code strictly decides execution.** Davis enforces this paradigm through low-level C-SIMD (AVX2/AVX-VNNI) kernel acceleration, static Abstract Syntax Tree (AST) interposition, fail-closed authorization bounds (SYS-021), post-quantum Merkle evidence ledgers, and zero-copy shared memory IPC spanning pocketable edge nodes and workstation hosts. In quantum information science, Davis has executed over 660,000 shots across physical IBM Heron r2 quantum processors, discovering statistically significant dynamical decoupling advantages and authoring pre-registered information recovery protocols.

Backed by comprehensive institutional credentials from the **Defense Acquisition University (DAU)** in Systems Engineering, Cybersecurity for Weapons Systems, DevSecOps, Middle Tier of Acquisition (Section 804), and Other Transaction Authorities (OTAs), Davis bridges low-level kernel physics with federal acquisition rigor.

CORE TECHNICAL COMPETENCIES

Domain	Technical Capabilities & Technologies
Low-Level Systems & Kernels	C/C++17, AVX2, AVX-VNNI, FMA Vector Intrinsics, Memory Stride Alignment, POSIX AF_UNIX Sockets, `epoll`, Android NDK JNI (`DirectByteBuffer` Zero-Copy Shared Memory), Linux Kernel IPC, Bash Automation.
Sovereign AI & Neural Runtimes	Decoupled Dual-Voice Architecture, Non-Causal Language Models (NCLM), `osiris.nclm` (4.08M param living core), PyTorch, Vertex AI SFT (`gemini-2.5-flash`), Ollama (`qwen2.5:1.5b`), MTEB/BEIR Benchmarking (`EXP-OLMO3-BEIR-SCIFACT-001`).
Quantum Information Science	IBM Quantum Heavy-Hex Processors (`ibm_fez`, `ibm_torino`, `ibm_marrakesh`), OpenQASM 3.0 AST Parsing, Staggered XY8/XY4 Dynamical Decoupling (STAG-XY8), Hayden–Preskill Scrambling (`osiris-hp-001`), Bell-CHSH Inequalities, PennyLane QML.
Cryptographic Governance	SYS-021 Zero Self-Supplied Authorization, NIST FIPS 204 ML-DSA-65 (Lattice PQC) + Ed25519 Hybrid Dual-Signing, Merkle Hash Chains (`evidence_ledger.db`), RFC 8785 Canonical JSON Serialization, In-Memory AST Claims Firewall (`claims.py`).
Distributed Edge Architecture	Multi-Node Mesh (ARM64 Google Pixel 10 Pro Fold / Android 16 / Termux $\leftrightarrow$ Workstation x86_64 WSL2 Ubuntu), Abstract Domain Sockets (`\0osiris_bt_bridge`), Tensor G5 Thermal Governors & 300s Execution Horizons.
Parametric CAD & Manufacturing	`osiris-fabric` Autonomous 12-Printer Bambu Lab Fleet (P1S, H2X, A1, A2L, A1 Mini), Multi-Material AMS Flush Optimization (ΔE 40%-70% Waste Reduction), Headless Cloud Run OpenSCAD/OrcaSlicer Geometry Compile Gates.
Defense Acquisition & Compliance	CMMC 2.0 Level 2/3 (NIST SP 800-171/800-53), Middle Tier of Acquisition (Section 804 / `ACQ 0150`), Other Transactions (`CLC 066`), Cybersecurity for Weapons Systems (`CYB 5610`), DevSecOps (`WSA 002`), Integrated Master Scheduling (`CLM 094`, Primavera P6).

PROFESSIONAL EXPERIENCE

Founder, Lead Systems Architect & Principal Researcher

Agile Defense Systems LLC · *Remote / Workstation Substrate* · **2022 – Present**

- **Sovereign AI Dual-Voice Engine:** Designed and deployed a decoupled neural computing architecture pairing a conversational mentor (`qwen2.5:1.5b`) with a localized 4.08M parameter living core (`osiris.nclm`), achieving sovereign offline execution with zero external cloud dependencies.
- **C-SIMD Vector Kernel Optimization:** Authored hand-tuned C-SIMD kernels (`c_simd_ops.c`) leveraging AVX2 and AVX-VNNI hardware flags. Enforced strict C-contiguous float32 memory stride alignment across the autoregressive (B, H, T, HD) KV-cache, eliminating segmentation faults and reducing forward-pass logit error to $< 1.15 \times 10^{-5}$ deviation from `SovereignTransformerV2`.
- **SYS-021 Cryptographic Governance:** Architected and implemented the SYS-021 Zero Self-Supplied Authorization standard, mandating that every state-modifying action requires an immutable cryptographic receipt containing the operator identity, timestamp, and exact SHA-256 payload digest ($\mathscr{H}_{\text{payload}}$ = SHA-256(code)). Any byte modification instantly invalidates execution grants.

- **Hybrid Post-Quantum Cryptographic Enclave:** Engineered dual-signing verification routines (`osiris_cli/pqc_enclave.py`) combining NIST FIPS 204 ML-DSA-65 (lattice-based digital signatures) with Ed25519, anchoring state transitions to an append-only SQLite Merkle ledger (`evidence_ledger.db`) formatted via RFC 8785 Canonical JSON.
- **Autonomous Manufacturing Control Plane (`osiris-fabric`):** Automated production across an active 12-printer Bambu Lab fleet (5x P1S, 1x H2X, 2x A1, 2x A2L, 2x A1 Mini). Developed `AmsFlushOptimizer` tracking digital spool twins and calculating color distance (ΔE), reducing multi-material purge waste by 40% to 70% via infill purge diversion.
- **Cloud Run CAD Geometry Gates:** Deployed containerized Cloud Run microservices executing headless OpenSCAD and OrcaSlicer CLI binaries. Enforced fail-closed geometry gates preventing unverified or non-manifold parametric scripts from compiling into machine G-code.
- **Normative Platform Acceptance (`osiris-android-spec-001`):** Formulated and verified the comprehensive 46-criterion Android platform specification across 10 functional modules, demonstrating 100% compliance across 53 unit, mocked, and physical device test suites.

Quantum Cyberinfrastructure & Information Science Lead

Independent Research / IBM Quantum Researcher Program · 2024 – Present

- **Physical QPU Campaign Execution:** Designed, scheduled, and verified quantum optimal control sequences across physical 156-qubit IBM Heron r2 heavy-hex superconducting QPUs (`ibm_fez`, `ibm_torino`, `ibm_marrakesh`), accumulating over 663,552 raw experimental shots.
- **Staggered Dynamical Decoupling (STAG-XY8):** Pioneered staggered XY8 and XY4 pulse schedules targeting spectator transmon qubit coupling chains (edges 14, 15 and 61, 62). Measured residual Hamiltonian ZZ interactions ($\mathscr{H}_{ZZ} = (\zeta_{uv} / 4) \cdot Z_u Z_v$) and demonstrated a statistically significant +0.0186 fidelity improvement over simultaneous unmitigated baselines ($p < 10^{-4}$).
- **Hayden–Preskill Quantum Recovery Protocol (`osiris-hp-001`):** Authored and executed a 3-arm pre-registered quantum information recovery experiment measuring EPR state decoding ($F_{rec} = \langle\Phi^+|\rho_{RM'}|\Phi^+\rangle$) across Identity, scrambling, and noise-calibrated hardware baselines.
- **In-Memory Claims Verification Engine:** Authored `claims.py`, an in-memory physical claims verification firewall. Evaluates transcompiled circuits against Bell-CHSH bounds, unitary conservation laws, and empirical hardware noise floors (1.11 pp RMS), automatically rejecting unphysical or speculative claims prior to execution.
- **OpenQASM 3.0 AST Parser:** Implemented an in-memory Abstract Syntax Tree parser and transpiler for OpenQASM 3.0 circuits, generating calibrated pulses natively mapped to Heron r2 ISA gates (`cz`, `rz`, `sx`, `x`).

Edge Substrate & Distributed Systems Engineer

Agile Defense Systems LLC · 2023 – Present

- **Distributed Dual-Node Substrate:** Built a robust low-latency computing mesh connecting pocketable ARM64 edge nodes (Node Alpha: Google Pixel 10 Pro Fold running Android 16 / Termux) with workstation hosts (Node Beta: x86_64 WSL2 Ubuntu 24.04 LTS).
- **Zero-Copy NDK JNI Bridge:** Bypassed Android Runtime (ART) garbage collection stalls by authoring C++ Android NDK shared memory interfaces (`NativeBleBridge`) utilizing `DirectByteBuffer` pointers.

- **Abstract Domain Socket IPC:** Implemented Linux abstract domain socket daemons (`\0osiris_bt_bridge`), delivering sub-150 microsecond POSIX `epoll` packet streaming between hardware BLE radios and host CLI tools.
- **Substrate Governor & Thermal Guard:** Engineered hysteretic hardware monitors (`osiris_daemon`) on Google Tensor G5 silicon, enforcing dynamic 300-second execution horizons to protect long-running tasks from Android Phantom Process Killers and kernel thermal throttling.

RIGOROUS EMPIRICAL EVIDENCE & PLATFORM BENCHMARKS

Evidence Capsule: `EXP-OLMO3-BEIR-SCIFACT-001` (MTEB / BEIR)

- **Scope:** Evaluated `allenai/Olmo-3-7B-Instruct` on the SciFact benchmark using an isolated local bounded sandbox execution policy within the `osiris-cybercontext` ecosystem.
- **Empirical Retrieval Metrics:**
- **NDCG@10:** `0.2658` (Normalized Discounted Cumulative Gain across top-10 retrieved documents).
- **MRR@10:** `0.1139` (Mean Reciprocal Rank at rank 10).
- **Corpus Task Complexity (CTC):** `0.7646` (Derived as CTC = 1 − (0.8 · NDCG@10 + 0.2 · MRR@10)).
- **Diagnostic Finding (Top-1 Rank Suppression):** Achieved 80% Recall@10 (ground-truth found in 4 of 5 queries), but discovered severe rank suppression where targets landed exclusively at ranks 6, 6, 8, and 9 (zero hits in ranks 1–5), capping MRR@10 at 0.1139 and proving the mathematical necessity of cross-encoder rerankers.
- **Cryptographic Anchors:** Merkle Tree Root: `e46e6d92da6abc1819a6d4c0c19d8f29ded830598465d760cf5d75755fdad5b6` · Evidence Bundle SHA-256: `8883af4cf09d8d504ff1eeff65e71c71c7a6a16067bf0c8a837828e9120f2ce6` (`benchmark_capsule.evidence.zip`).

Normative Acceptance Specification: `osiris-android-spec-001`

- **Test Verification Status:** 53 of 53 tests passed (100% pass rate; zero failures) across 10 core Android modules (`:demo` , `:app` , `:keyboard` , `:governance` , `:evidence` , `:evaluation` , `:execution` , `:runtime-bridge` , `:sdk`).
- **Enforced Standards:** CIL L0–L8 security barriers, direct memory pointer alignment, zero-copy socket streaming, and unalterable SQLite hash ledger persistence.

PERMANENT SCIENTIFIC PRE-REGISTRATIONS & ZENODO DOIS

All scientific findings, QPU telemetry runs, quantum optimal control scripts, and software releases are permanently preserved in the open scientific record with immutable Digital Object Identifiers (DOIs):

Investigation / Deliverable Title	Scientific Scope & Description	Permanent DOI Link
osiris-hp-001 Hayden–Preskill Protocol	Pre-registered 3-arm EPR scrambling and quantum recovery benchmark on physical IBM Heron r2.	doi:10.5281/zenodo.23968569
Staggered XY4 Decoupling Controlled Tests	Pre-registered controlled test campaigns evaluating 8-qubit heavy-hex coupling chains on `ibm_fez` & `ibm_marrakesh`.	doi:10.5281/zenodo.22855102
IBM QPU Multi-Node Telemetry Dataset	Raw execution telemetry and calibration records across 663k shots on Heron r2 processors.	doi:10.5281/zenodo.22870287
NCLM Living Core Gate Pilot-0 Evaluation	Empirical evaluation scoring living core bits-per-byte (bpb) performance on held-out exchanges.	doi:10.5281/zenodo.23075229
osiris CyberContext Companion Runtime	Published benchmark runtime and evaluation suite for AI context verification and retrieval evaluation.	doi:10.5281/zenodo.23305204
osiris Architecture v5.3.0 Release Package	Complete cryptographic release capsule, C-SIMD sources, and SQLite hash-ledger implementation.	doi:10.5281/zenodo.23880340
osiris Architecture v5.3.1 Release Package	Patch release with enhanced PQC ML-DSA-65 dual-signing and edge daemon thermal governors.	doi:10.5281/zenodo.23884323

DEFENSE ACQUISITION UNIVERSITY (DAU) CREDENTIALS

Official transcripts verified under Learner Record: Devin Davis (Universal Profile).

Cybersecurity, Cloud & Modern DevSecOps

- **CYB 5610:** Cybersecurity & Resiliency for Weapons, Control and IT Systems *(Completed 10/12/2026)*
- **CENG 003A:** AI Foundations for the DoD Introduction *(Completed 08/06/2024)*
- **WSA 002:** DEVSECOPS for the DoD: Security Focus *(Completed 03/22/2023)*
- **CLE 075:** Introduction to DoD Cloud Computing *(Completed 07/09/2023)*
- **ACQ 1300:** Fundamentals of Technology Security / Transfer *(Completed 11/10/2022)*
- **CLE 076:** Introduction to Agile Software Acquisition *(Completed 05/21/2022)*
- **CLE 077:** Defense Business Systems (DBS) Acquisition *(Completed 11/16/2022)*

Systems Engineering, R&D & Technology Prototyping

- **ENG 201:** Applied Systems Engineering in Defense Acquisition, Part I *(Completed 06/19/2022)*
- **ENG 101:** Fundamentals of Systems Engineering *(Completed 06/10/2022)*
- **CON 0210:** R&D Processes & Programs *(Completed 06/24/2022)*
- **CCON 021A:** Legal Considerations for Research and Development Instruments *(Completed 01/30/2023)*
- **ACQ 0800:** Federally Funded Research and Development Centers (FFRDCs) *(Completed 01/18/2023)*
- **Requirements Engineering: A Good Practice Guide** *(Completed 11/16/2022)*

Rapid Acquisition, Middle Tier (MTA) & Other Transactions (OTAs)

- **ACQ 0150:** Middle Tier of Acquisition (Section 804 Rapid Prototyping/Fielding) *(Completed 01/17/2023)*
- **CLC 066:** Other Transactions (OTs) *(Completed 06/11/2022)*
- **ACQ 1010:** Fundamentals of Systems Acquisition Management *(Completed 01/18/2023)*
- **ETM 1090:** Technical Perspectives on Defense Contracting Fundamentals *(Completed 01/13/2023)*
- **CON 0160:** Competition in Contracting *(Completed 01/18/2023)*
- **CON 124:** Contract Execution *(Completed 05/28/2022)*
- **CON 0820:** Critical Concepts in Operational Contract Support *(Completed 07/13/2023)*
- **FAC 086:** Back to Basics: Awarding Task Orders Under Multiple-Award IDIQ Contracts *(Completed 12/13/2022)*
- **CLC 002:** Sole Source Streamlining ToolBox (SSST) *(Completed 01/18/2023)*
- **FAC 027:** GSA Schedules and Small Business *(Completed 01/06/2019)*
- **ACQ 165:** Defense Acquisition of Services *(Completed 01/22/2019)*

Integrated Master Scheduling, Cost Analysis & Project Governance

- **CLM 094:** Integrated Master Scheduling (IMS) *(Completed 03/17/2023)*
- **Primavera P6 Essential Training** *(Completed 03/20/2023)*
- **Managing Resource-Constrained Projects with Microsoft Project** *(Completed 03/19/2023)*
- **CLC 056:** Analyzing Contract Costs *(Completed 11/17/2022)*
- **CLB 037:** Defense Working Capital Fund *(Completed 12/13/2022)*
- **CLC 043:** Defense Priorities and Allocations System *(Completed 01/28/2023)*
- **CCON 002:** Importance of Business Acumen *(Completed 05/24/2022)*
- **CLM 049:** Procurement Fraud Indicators *(Completed 01/18/2023)*
- **CLC 132:** Organizational Conflicts of Interest *(Completed 01/21/2023)*
- **CLC 044:** Alternative Dispute Resolution *(Completed 11/08/2022)*
- **CLC 045:** Partnering *(Completed 08/04/2022)*
- **IND 105:** Contract Property Fundamentals *(Completed 02/07/2019)*
- **CLE 032:** Sustainable Manufacturing for DoD – Part 1 *(Completed 01/07/2019)*
- **CLC 046:** DoD Sustainable Procurement Program *(Completed 01/07/2019)*

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